Domain-Specialized Embedding Learning for Medical Retrieval-Augmented Generation (RAG)

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Introduction

- **Large Language Models (LLMs)**
  - Strong capabilities in natural language understanding and generation
  - Increasingly used in biomedical QA and clinical NLP
- **Clinical challenge**
  - LLMs can produce **hallucinated** or **factually incorrect responses**
  - High-stakes domains require **recent**, accurate, evidence-grounded outputs
- **The RAG solution**
  - Retrieval Augmented Generation (RAG) reduces hallucinations by grounding answers in verified external knowledge
- **The Bottleneck**
  - RAG is only as good as the document it retrieves.
  - If retrieval fails, the entire system fails

Problem Statement

- **The Problem:**
  - Most RAG systems rely on generic embedding models that are not optimized for complex biomedical language
  - This leads to:
  - Poor understanding of clinical terminology and abbreviations
  - Weak semantic matching between user queries and medical texts
  - Lower retrieval relevance
- **The cascading effect:**
  - **Low-quality retrieval → low-quality generation** (even with a strong LLM).
  - The result is less faithful, less trustworthy medical answers

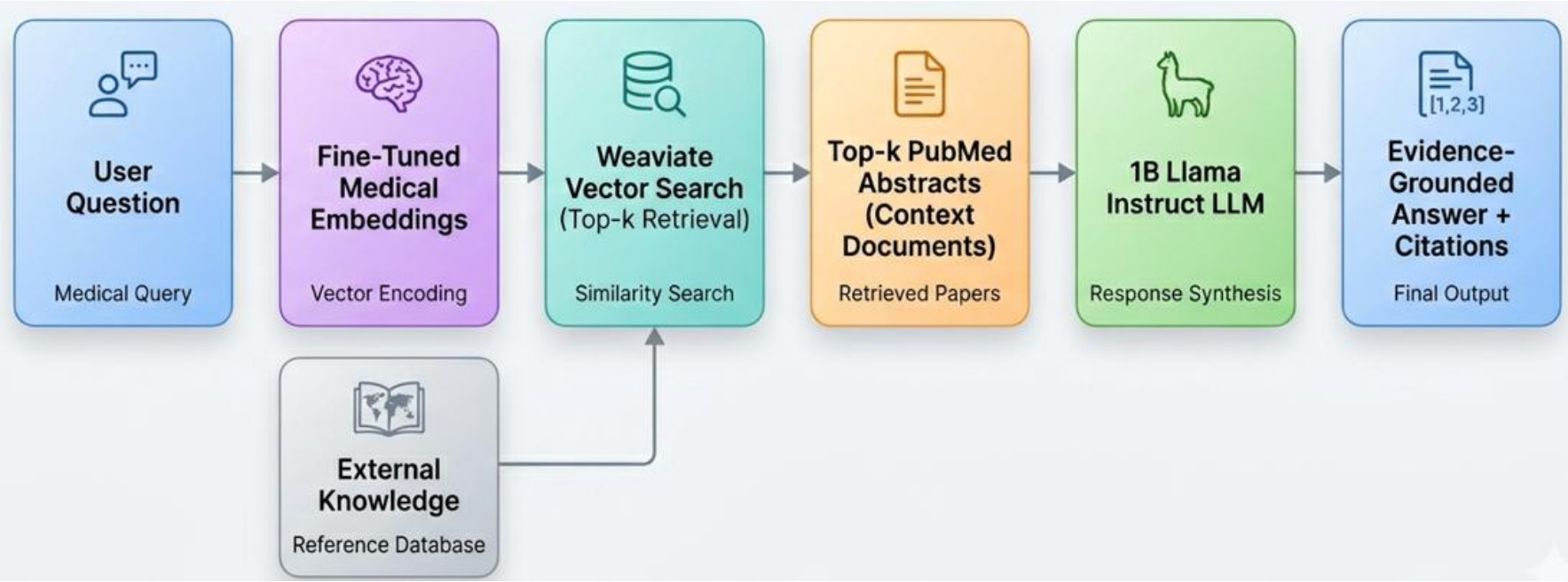
Can domain-specific embedding models significantly improve retrieval quality and reduce hallucinations in medical Retrieval-Augmented Generation (RAG) systems?

What is RAG?

Retrieval Augmented Generation (RAG) is a framework that combines

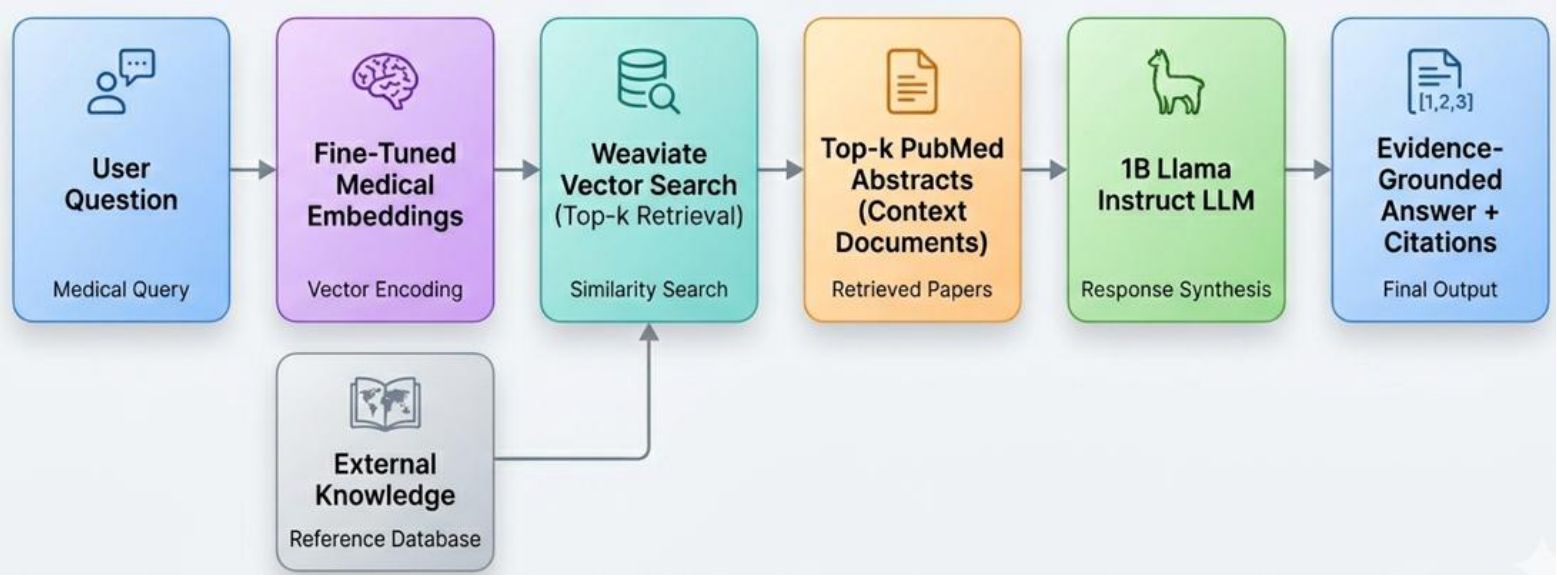
- ***Information retrieval***
- ***Language generation***

Rather than relying solely on internal memory, the model generates responses grounded in external evidence.



System Architecture

- **External Knowledge:** The external knowledge files are chunked, indexed and loaded to vector database
- **User query:** A medical question is entered by the user.
- **Embedding Model:** Converts query into a dense vector using our fine-tuned medical embeddings.
- **Vector Search (Weaviate):** Performs similarity search to retrieve top-k relevant PubMed abstracts.
- **Context Assembly:** High-quality biomedical evidence used as context.
- **LLM Synthesis:** LLM processes the context to generate an evidence-grounded answer.

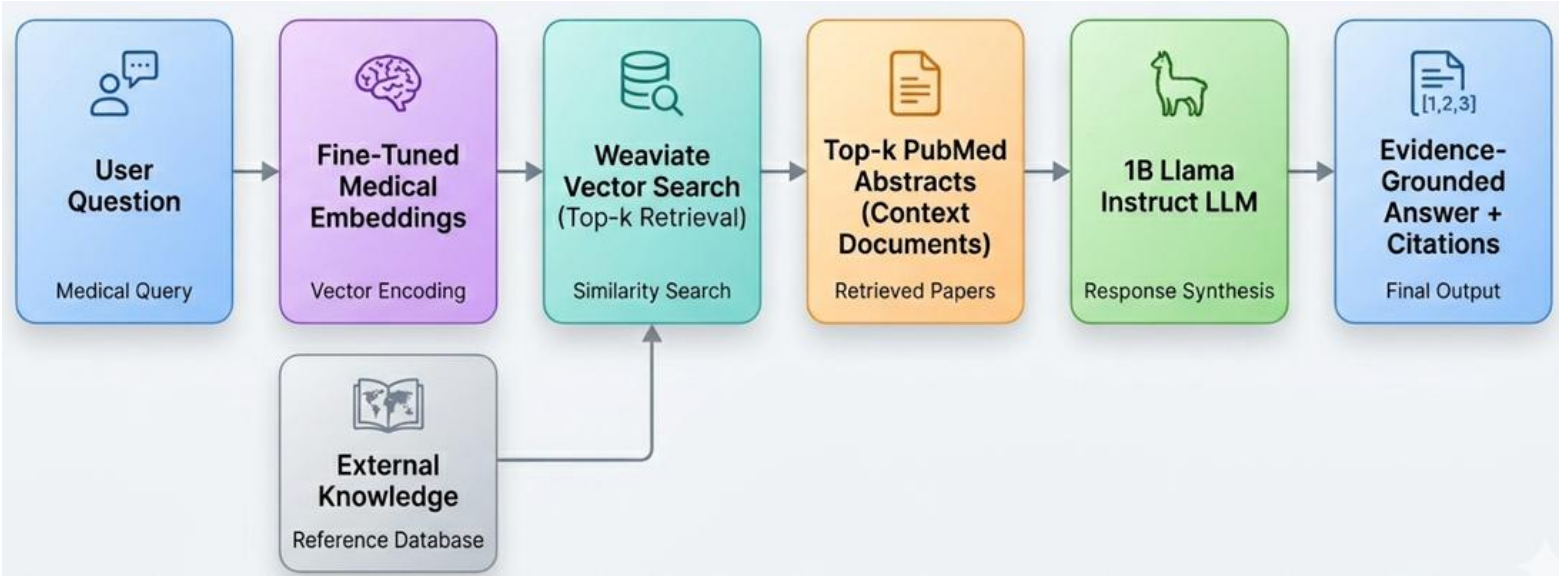


Components of RAG

Vector Database (Weaviate)
- Store document embeddings
- Enables fast semantic search

Retriever (Embedding Model)
- Searches for relevant documents
- Uses semantic similarity
- Returns top-k similar content

Generator (LLM)
- Ingests the query and retrieved context
- Performs reasoning over evidence to generate a grounded response



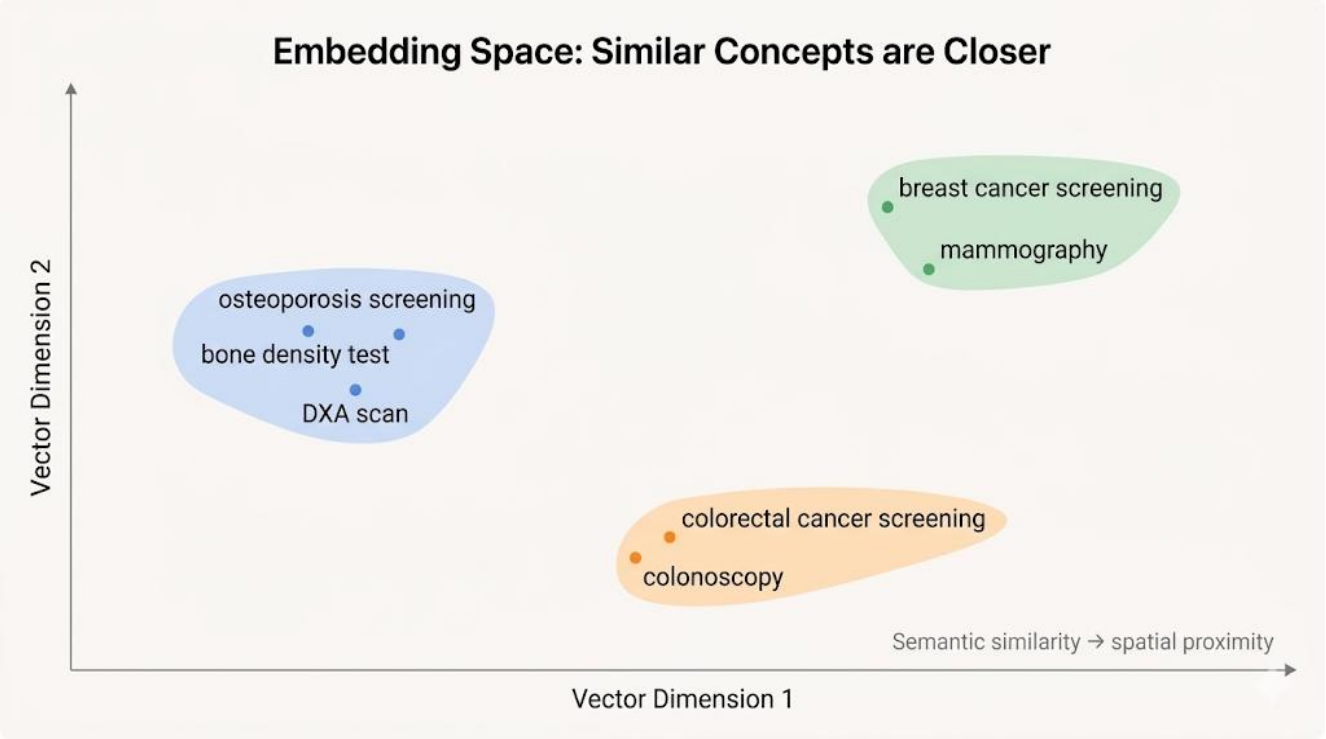
Embedding model

What is the purpose of an embedding model?

- Convert text to dense vectors
- Places ***similar concepts closer in the vector space***
- Captures semantic meaning
- Used for retrieval and similarity search

How is it different from the LLM?

- Embedding model → ***Search*** brain
- LLM → ***Reasoning*** brain



Fine-tuning Embedding model

What is Contrastive Learning?

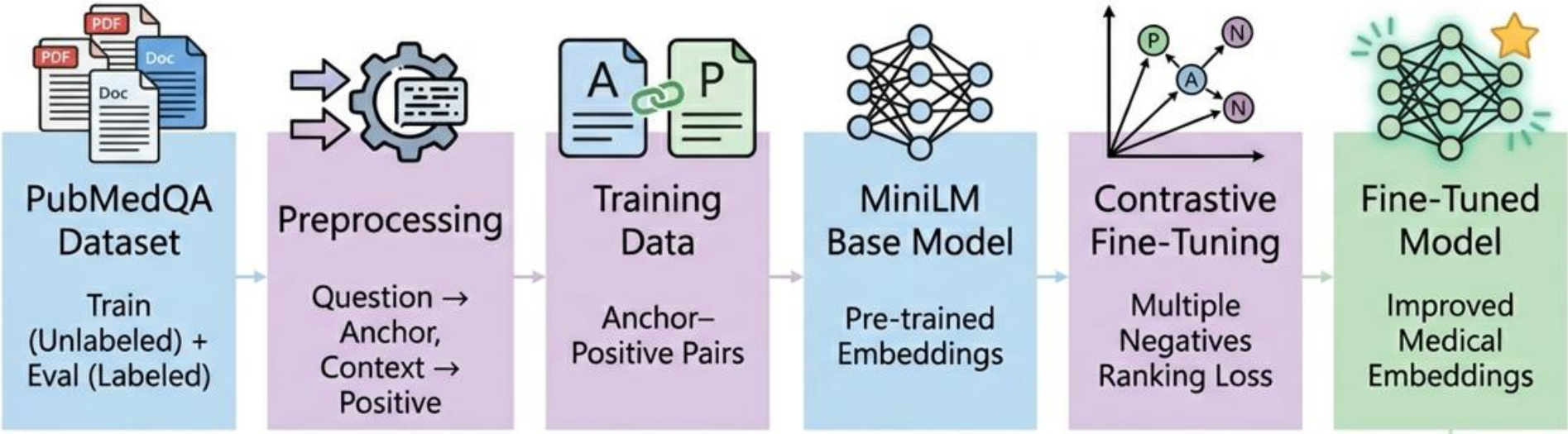
- A **representation learning technique** that trains models to distinguish between similar and dissimilar data samples.

Training Objective:

- **Pull similar pairs closer** in embedding space (Question + correct evidence)
- **Push dissimilar pairs apart** (Question + unrelated evidence)
- Implemented using InfoNCE Loss (Multiple Negatives Ranking Loss) to efficiently use all other abstracts in a batch as negative examples.

Result:

- Enables models to learn complex medical **semantic relationships** without explicit labels.



Experiment Setup

- **Dataset:**
  - *For finetuning*: PubMedQA – unlabeled split (10,000 samples)
  - *For evaluation*: PubMedQA – Labeled split (1000 samples)
- **Embedding models compared:**
  - *Generic Baseline:* all-MiniLM-L12-v2
  - *Domain-specialized:* Our custom mini-pubmedqa-finetuned
- **Generator:** Llama-3.2-1B-Instruct (prompted for strict Yes/No/Maybe outputs).
- **Infrastructure:**
  - *Weaviate Cloud (Vector DB)*
  - *SentenceTransformers*
  - *LangChain*
  - *Google Colab (T4 GPU)*
  - *Gradio UI*

Accuracy Evaluation

Overall Accuracy

Embedding Model	Accuracy	Correct	Total
all-MiniLM-L6-v2	54.50%	545	1000
mini-pubmedqa-finetuned	54.70%	547	1000

Class-Wise Performance:

Class	all-MiniLM-L6-v2	mini-pubmedqa-finetuned
Yes	58.33% (322/552)	56.70% (313/552)
No	65.98% (223/338)	68.93% (233/338)
Maybe	0.00% (0/110)	0.91% (1/110)

Conclusion

- Built a complete, end-to-end **Medical RAG system.**
- Successfully fine-tuned medical embeddings using contrastive learning.
- Integrated **Weaviate vector search** to retrieve verified PubMed evidence.
- Compared **generic vs. domain-fine-tuned embeddings.**
- Generated **evidence-grounded answers with citations.**

Future Work

- **Hard Negative Mining:** Train the embedding model with closely related but incorrect abstracts (hard negatives) to force the model to learn finer clinical distinctions.
- **Advanced RAG Architectures:** Implement Cross-Encoder Re-ranking models to re-order retrieved documents for even higher precision.
- **Real time clinical Expand the Dataset:** Train on broader, highly diverse clinical corpora such as MIMIC-III (electronic health records) or full-text PubMed articles.
- **Rich Answer Generation:** Expand the LLM prompt to generate detailed, heavily cited clinical summaries rather than simple Yes/No/Maybe outputs.

Demo

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